3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Logic diagrams. The logic diagrams shall be as specified on figure 2.

3.2.3 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 3.

3.2.4 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained and available upon request.

3.2.5 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended ambient operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.6.1. Total dose radiation hardness identifier. Total dose radiation hardness identifier shall be in accordance with MIL-M-38510 and 4.5.5 herein.

3.6.2 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

3.6.3 Correctness of indexing and markings. All devices shall be subjected to the final electrical tests specified in table II after part number marking to verify that they are correctly indexed and identified by part number. Optionally, an approved electrical test may be devised especially for this requirement.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 37 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Delete the sequence specified in 3.1.9 through 3.1.13 of method 5004 and substitute lines 1 through 7 of table II herein.

b. Burn-in (method 1015 of MIL-STD-883).

(1) Static tests (test condition A) using circuit shown on figure 4 or equivalent. Ambient temperature (T_A) shall be 125°C minimum. Test duration for each static test shall be 24 hours minimum for class S devices and in accordance with table I of method 1015 for class B devices.

(2) Dynamic test (test condition D) using circuit shown on figure 5 or equivalent. Ambient temperture (T_A) shall be 125°C minimum. Test duration shall be in accordance with table I of method 1015.

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c. Interim and final electrical parameters shall be as specified in table II herein.

d. For class S devices, post dynamic burn-in, or class B devices, post static burn-in, electrical parameter measurements may, at the manufacturer's option, be performed separately or included in the final electrical parameter requirements.

4.2.1 Percent defective allowable (PDA).

a. The class S devices PDA shall be 5 percent for static burn-in and 5 percent for dynamic burn-in, based on the exact number of devices submitted to each separate burn-in.

b. Static burn-in I and II failures shall be cumulative for determining PDA.

c. The class B devices PDA shall be in accordance with MIL-M-38510 for static burn-in. Dynamic burn-in is not required.

d. Those devices whose measured characteristics after burn-in exceed the specified delta (Δ) limits or electrical parameter limits specified in table III, subgroup 1, are defective and shall be removed from the lot. The verified failures divided by the total number of devices in lot initially submitted to burn-in shall be used to determine the percent defective for the lot and the lot shall be accepted or rejected based on the specified PDA.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D and E inspections (see 4.4.1 through 4.4.5).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.5).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be performed in accordance with table II herein.

b. Subgroups 5, 6, 7, and 8 of table I of method 5005 of MIL-STD-883 shall be omitted.

c. Subgroup 4 (C_I measurement) shall be measured only for initial qualification and after process or design changes which may affect input capacitance. Capacitance shall be measured between the designated terminal and V_{SS} at a frequency of 1 MHz.

d. Subgroup 12 shall be added to the group A inspection requirements for class S devices using an LTPD of 15 and consist of the procedures, test conditions, and limits specified in table III.

e. At the manufacturer's option, test tapes may be programmed simultaneously for each identical section provided that each output is measured and each specified input combination is tested.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. Class S steady state life (accelerated) test circuits shall be submitted to the qualifying activity for approval. When the alternate steady state life test is used, the circuit on figure 5, or equivalent, shall be used.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/ V_{SS} = 0 V, $-55°C \le T_A \le +125°C$, Unless otherwise specified		Device type	Limits Min	Limits Max	Unit
Positive clamping input to V_{DD}	$V_{IC(pos)}$	T_A = 25°C, V_{DD} = GND, V_{SS} = open, output = open, I_{IN} = 1 mA		01, 02, 03	---	1.5	V
Negative clamping input to V_{SS}	$V_{IC(neg)}$	T_A = 25°C, V_{DD} = open, V_{SS} = GND, output = open, I_{IN} = -1 mA		All	---	-6	V
Quiescent supply current	I_{SS} or I_{DD}	V_{DD} = 18 V, V_{IN} = V_{SS} or V_{DD}, All input combinations		All	---	-750	nA
High level output voltage	V_{OH1}	V_{DD} = 15 V, I_O ≤1 µA	Any one input = V_{SS}, All other inputs = V_{DD}	01	14.95	---	V
			V_{IN} = V_{SS}	03			
			V_{IN} = V_{DD}	04			
Low level output voltage	V_{OL1}	V_{DD} = 15 V, I_O ≤1 µA	Any one input = V_{DD}, All other inputs = V_{SS}	01	---	0.05	V
			V_{IN} = V_{DD}	02, 03			
			All enables = V_{DD}, All other inputs = V_{SS}	04			
Input high voltage	V_{IH1}	V_{DD} = 5 V, see table III, I_O ≤1 µA		All	3.5	---	V
	V_{IH2}	V_{DD} = 10 V, see table III, I_O ≤1 µA		All	7.0	---	V
	V_{IH3}	V_{DD} = 15 V, see table III, I_O ≤1 µA		All	11.0	---	V

See notes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ V_{SS} = 0 V, -55°C ≤ T_A ≤ +125°C, Unless otherwise specified		Device type	Limits Min	Limits Max	Unit
Input low voltage	V_{IL1}	V_{DD} = 5 V, see table III, I_O ≤1 µA		All	---	1.5	V
	V_{IL2}	V_{DD} = 10 V, see table III, I_O ≤1 µA		All	---	3.0	V
	V_{IL3}	V_{DD} = 15 V, see table III, I_O ≤1 µA		All	---	4.0	V
Output high (source) current	I_{OH1}	V_{DD} = 5 V, V_{IN} = V_{SS}, V_{OH} = 4.6 V		01 03	-0.36	---	mA
	I_{OH2}	V_{DD} = 15 V, V_{OH} = 13.5 V	V_{IN} = V_{SS} V_{IN} = V_{DD}	01,03 04	-2.4	---	mA
Output low (sink) current	I_{OL1}	V_{DD} = 5 V, V_{OL} = 0.4 V	V_{IN} = V_{DD} V_{IN} = V_{SS}	02 01 03	12 0.36	--- ---	mA
	I_{OL2}	V_{DD} = 15 V, V_{OL} = 1.5 V	V_{IN} = V_{DD} V_{IN} = V_{SS}	02 01 04 03	80 2.4	--- ---	mA
Input leakage 2/ current	I_{IH}	V_{DD} = 18 V, Measure inputs sequentially; connect all unused inputs to V_{SS}		All	---	45	nA
Input leakage 2/ current	I_{IL}	V_{DD} = 18 V, Measure inputs sequentially; connect all unused inputs to V_{DD}		All	---	-45	nA
Input capacitance	C_{IN}	V_{DD} = 0 V, f = 1 MHz, T_A = 25°C		All	---	7.5	pF
Input test voltage	V_{ZAP}	C_1 = 100 pF, R_Z = 1.5 kΩ (see 4.5.3)		All	400	---	V

See notes at end of table.

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TABLE II. Electrical test requirements.

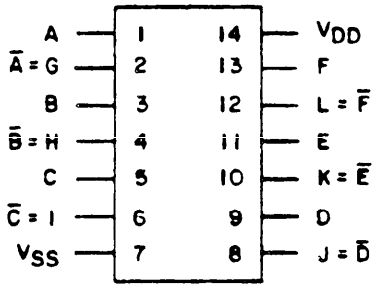
Line no.	MIL-STD-883 test	Class S device 3/				Class B device 3/			
		Ref. par.	Table 2/ III subgroups	Table 1/ IV delta limits	Test circuit figure	Ref. par.	Table 2/ III subgroups	Table 1/ IV delta limits	Test circuit figure
1	Interim electrical parameters method 5004		1				1		
2	Static burn-in I method 1015	4.2b 4.5.2			4				
3	Same as line 1		1	Δ					
4	Static burn-in II method 1015	4.2b 4.5.2			4	4.2b 4.5.2	4/		4
5	Same as line 1		1*	Δ		4.2d	1*	Δ	
6	Dynamic burn-in method 1015	4.2b 4.5.2			5				
7	Same as line 1	4.2d	1*	Δ					
8	Final electrical parameters method 5004		1*,2,3,9				1*,2,3,9		
9	Group A test requirements method 5005	4.4.1	1,2,3,4, 9,10,11			4.4.1	1,2,3, 4,9		
10	Group B end point electrical parameters method 5005	4.4.2	1,2,3, 9,10,11	Δ					
11	Group C end point electrical parameters method 5005					4.4.3	1,2,3	Δ	
12	Additional group C tests method 5005					4.4.3c	10,11		
13	Group D end point electrical parameters method 5005	4.4.4	1,2,3			4.4.4	1,2,3		

NOTES:
1/ (Δ) indicated delta limit shall be required only on table III subgroup 1, where specified, and delta values shall be computed with reference to the device interim electrical measurements.
2/ (*) indicates PDA applied to subgroup 1 (see 4.2.1).
3/ Blank spaces indicates tests are not applicable.
4/ The device manufacturer may at his option either perform delta measurements or within 24 hours after burn-in (or removal of bias) perform the final electrical parameter measurements.

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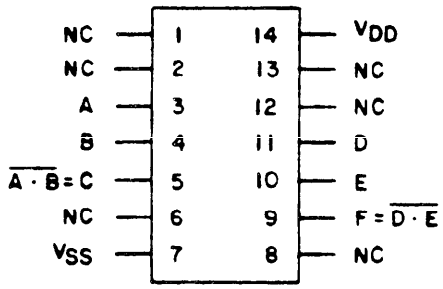
Device type 01

Cases A, C, D, X and Y



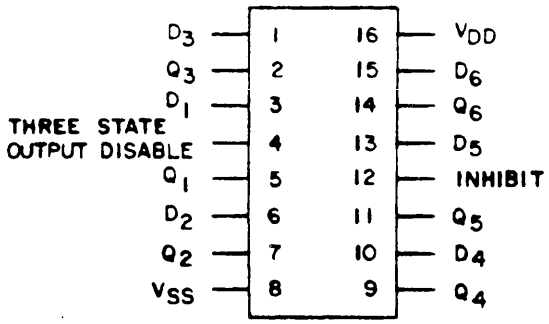
Device type 02

Cases A, C, D, X and Y



Device type 03

Cases E, F and Z



Device type 04

Cases E, F and Z

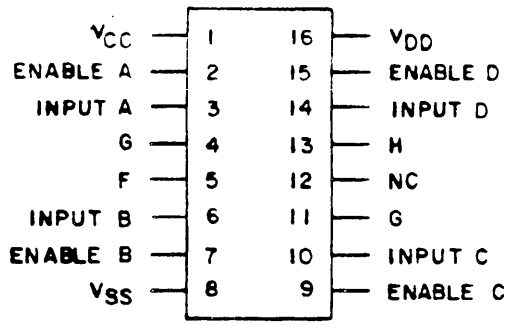
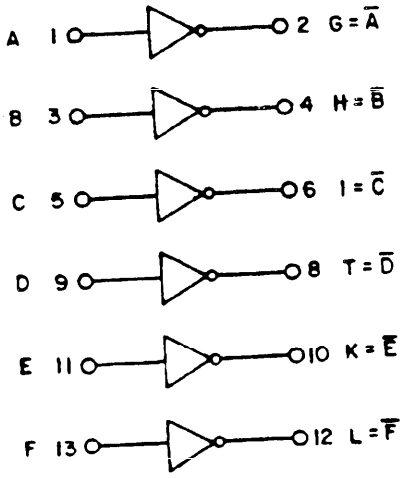


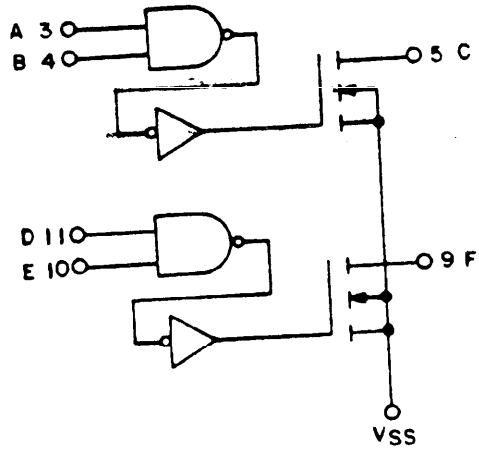
FIGURE 1. Terminal connections.

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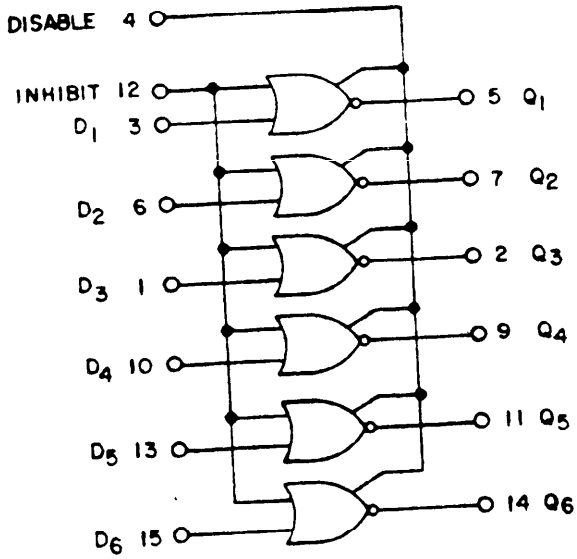
Device type 01



Device type 02



Device type 03



Device type 04

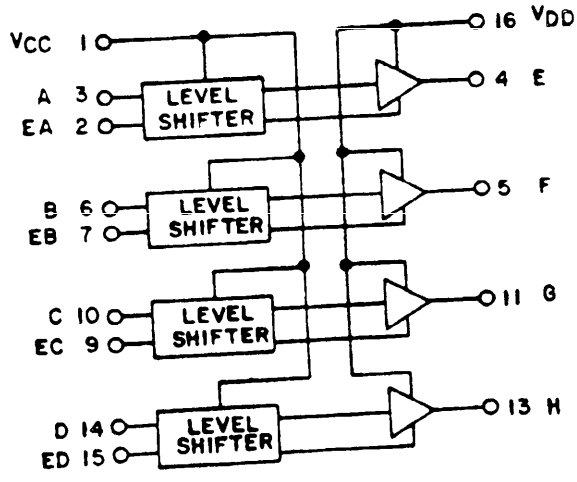


FIGURE 2. Logic diagrams.

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Device type 01

Truth table, each inverter	
Input	Output
L	H
H	L

Device type 02

Truth table, each gate		
Inputs		Outputs
A	B	
L	L	H
L	H	H
H	L	H
H	H	L

Device type 03

Truth table, each gate			
Inputs			Outputs
Disable	Inhibit	D_{IN}	Q_N
L	L	L	H
L	L	H	L
L	H	X	L
H	X	X	Z

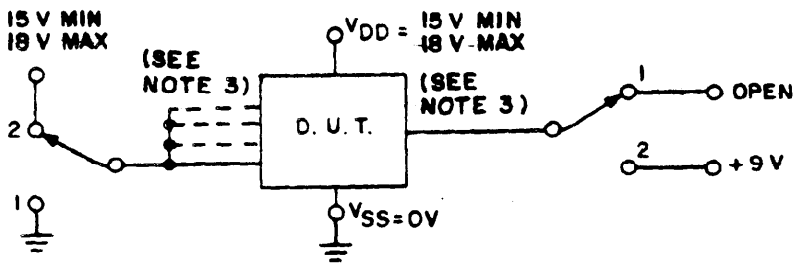
Device type 04

Truth table, each gate			
Inputs			Outputs
Mode	Enable		
L to H level shift	H	L	L
	H	H	H
	L	X	Z

NOTES:
1. Z = High impedance state
2. X = Don't care
3. H = High
4. L = Low
5. Device type 04 only; L = V_{SS}, H = V_{CC} at inputs and V_{DD} at outputs.

FIGURE 3. Truth tables.

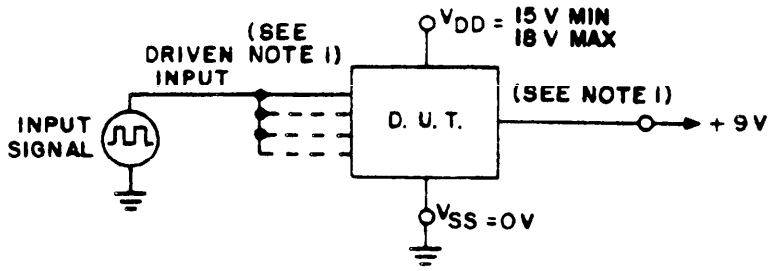
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NOTES:
1. For static burn-in I, all inputs are connected to 0 volts, switch position 1.
2. For static burn-in II, all inputs are connected to V_{DD}, switch position 2.
3. Except for V_{DD} and V_{SS}, each terminal shall be connected through a resistor whose value is 2 kΩ to 47 kΩ. The actual measured value of the resistor selected shall not exeed ±20% of its branded value due to use, heat or age.
4. Output may be in switch position 1 or 2.

FIGURE 4. Static burn-in test circuit.

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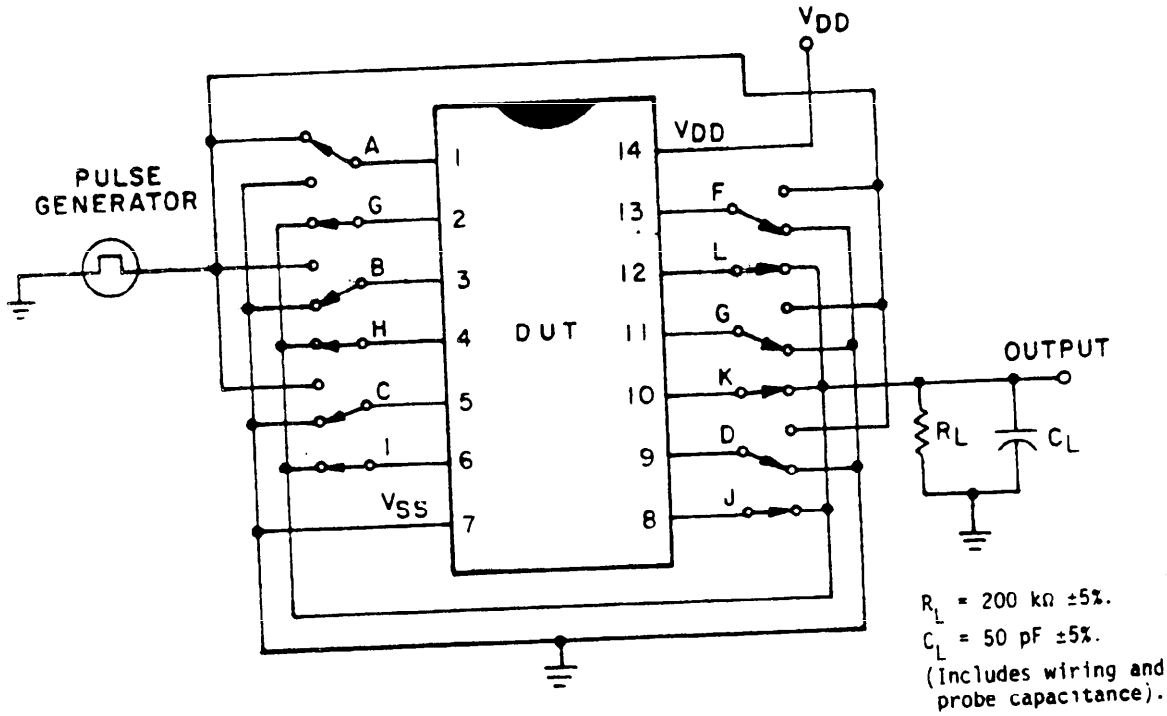


NOTES:
1. Except for V_{DD} and V_{SS}, each terminal shall be connected through a resistor whose value is 2 kΩ to 47 kΩ. The actual measured value of the resistor selected shall not exceed ±20% of its branded value due to use, heat or age.
2. Input signal requirements:
   a. Square wave, 50% duty cycle.
   b. 25 kHz < PRR < 1 MHz.
   c. t_{TLH} and t_{THL} < 1 µs.
   d. Voltage level:
      Minimum = V_{SS} -0.5 V, +10% V_{DD}.
      Maximum = V_{DD} +0.5, -10% V_{DD}.
   e. For device type 04; pin 4 (output disable) = PRR; all other inputs = PRR/2.

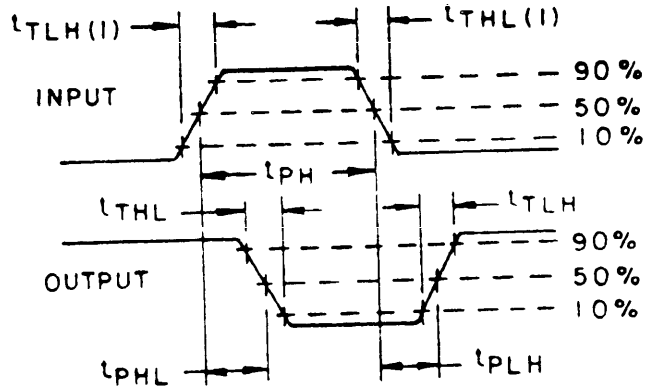
FIGURE 5. Dynamic burn-in and steady-state life test circuits.

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Device type 01



R_L = 200 kΩ ±5%.
C_L = 50 pF ±5%.
(Includes wiring and probe capacitance).



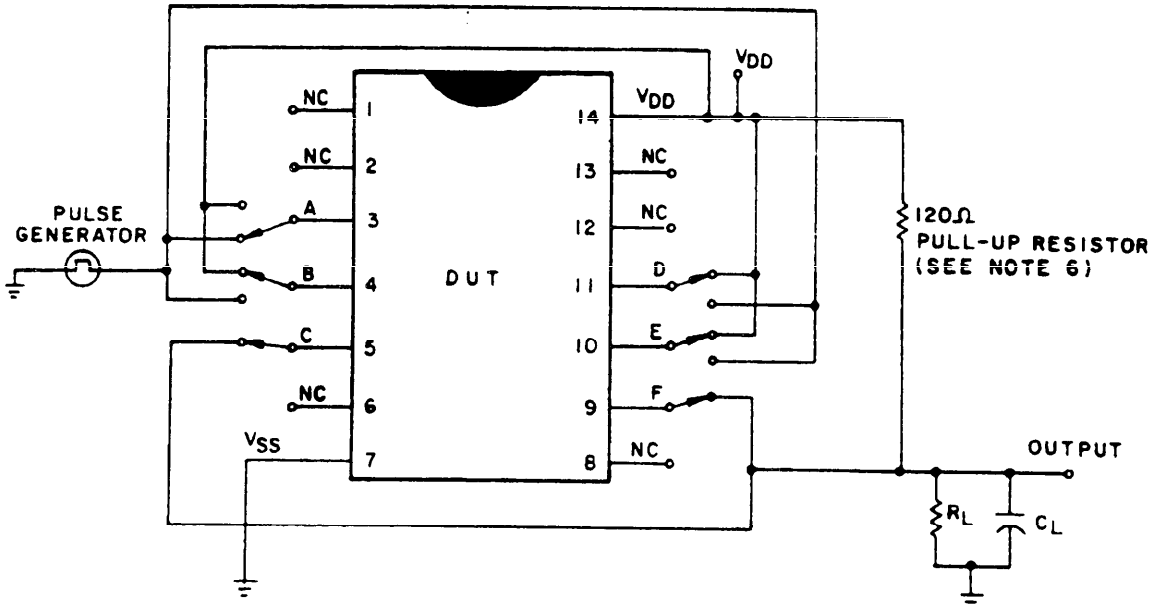
Dynamic Test Waveforms

Input pulse
$V_{GEN} = V_{DD}$ ±1.0%
t_{PH} = 1.0 ±0.1 µs
$t_{THL(1)} = t_{TLH(1)}$ = 20 ±2 ns
PRR = 200 kHz

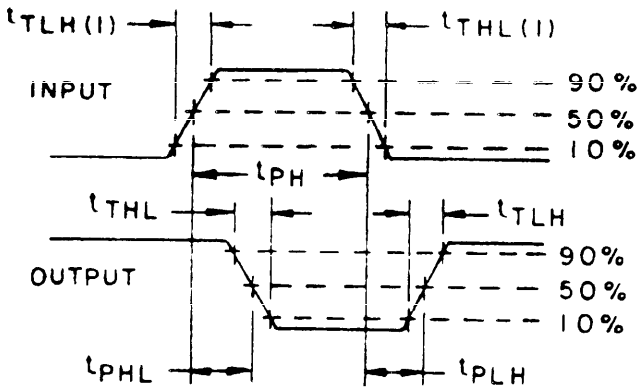
FIGURE 6. Switching time test circuit and waveforms

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Device type 02



R_L = 200 kΩ ±5%.
C_L = 50 pF ±5%.
(Includes wiring and probe capacitance).



DYNAMIC TEST WAVEFORMS

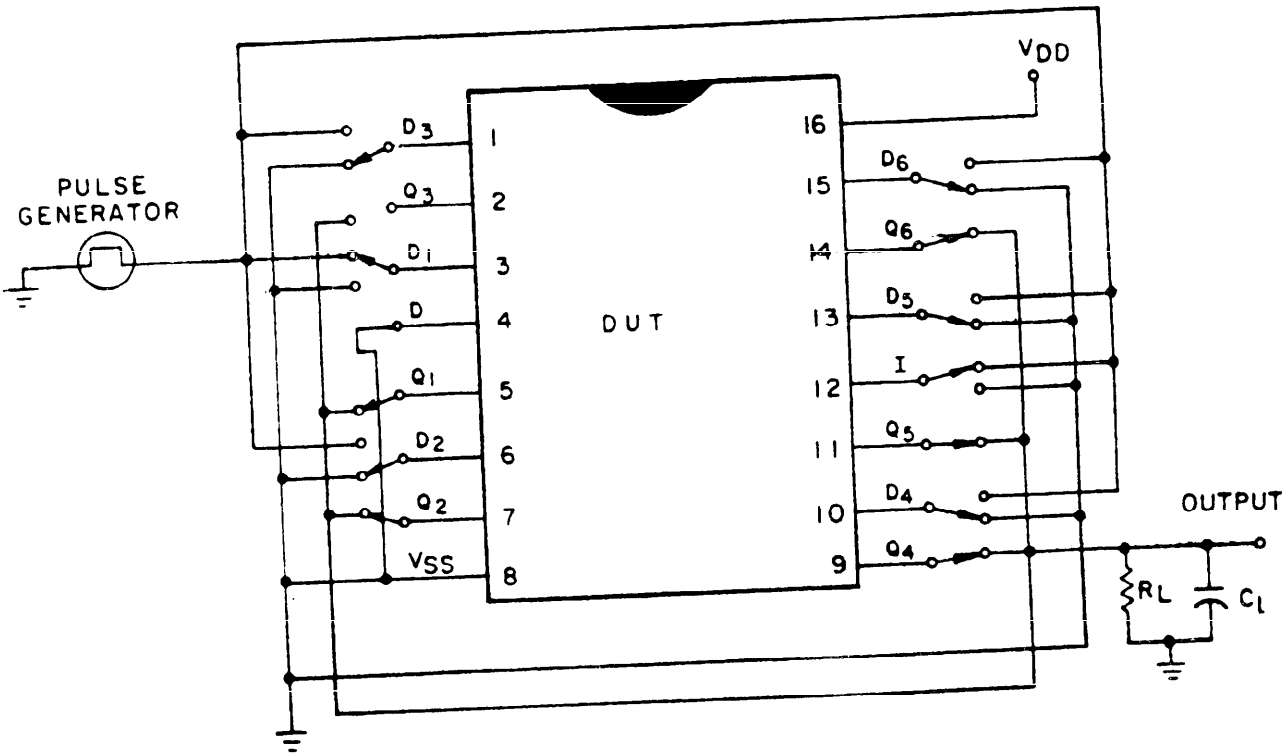
Input pulse

$V_{GEN} = V_{DD}$ ±1.0%
t_{PH} = 1.0 ±0.1 µs
$t_{THL(1)} = t_{TLH(1)}$ = 20 ±2 ns
PRR = 200 kHz

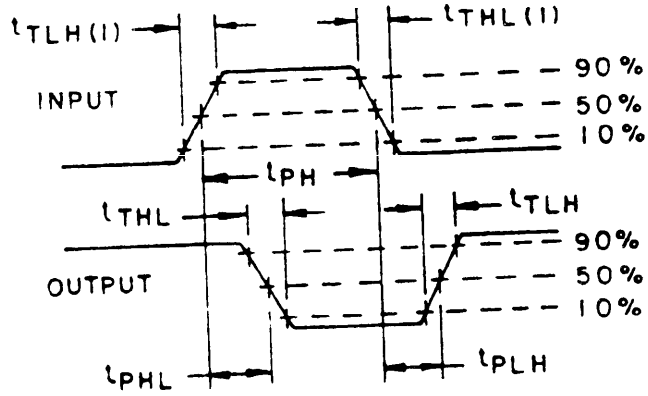
FIGURE 6. Switching time test circuit and waveforms - Continued.

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Device type 03



R_L = 200 kΩ ±5%.
C_L = 50 pF ±5%.
(Includes wiring and probe capacitance).



DYNAMIC TEST WAVEFORMS

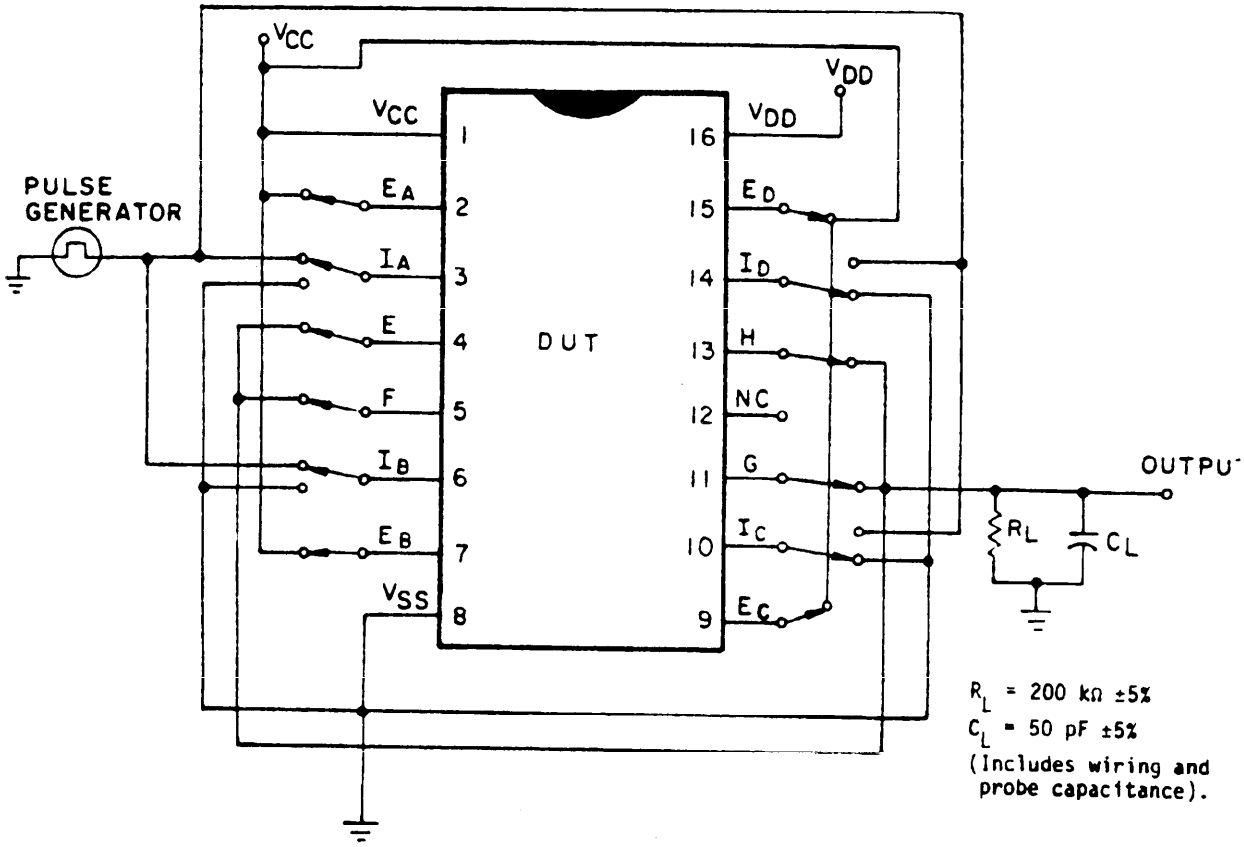
Input pulse

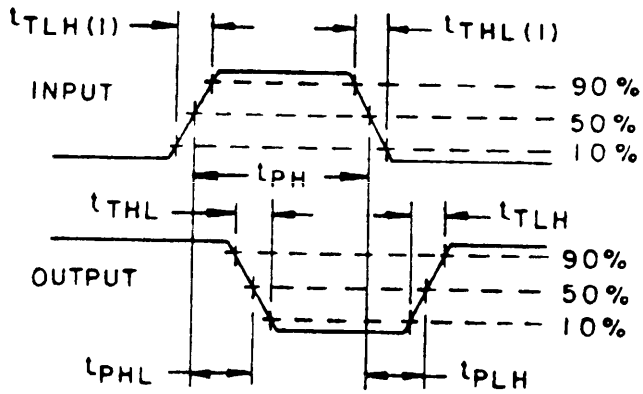
$V_{GEN} = V_{DD}$ ±1.0%
t_{PH} = 1.0 ±0.1 µs
$t_{THL(1)} = t_{TLH(1)}$ = 20 ±2 ns
PRR = 200 kHz

FIGURE 6. Switching time test circuit and waveforms - Continued.

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Device type 04





DYNAMIC TEST WAVEFORMS

Input pulse

$V_{GEN} = V_{DD} \pm 1.0\%$

$t_{PH} = 1.0 \pm 0.1$ µs

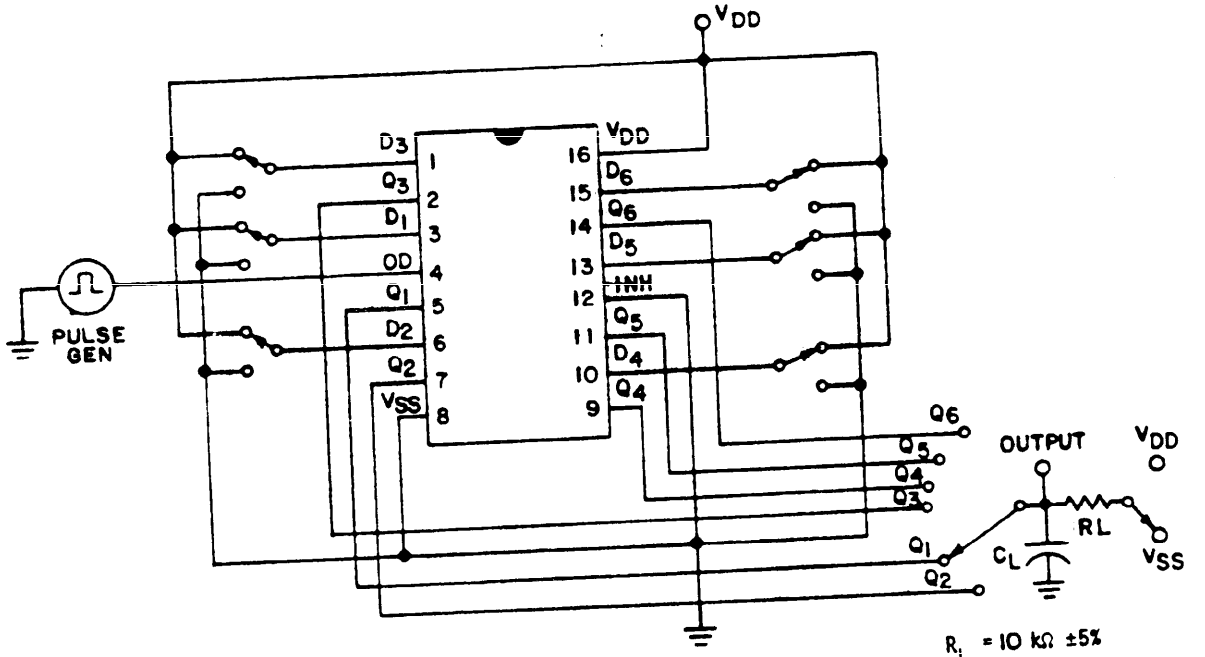
$t_{THL(1)} = t_{TLH(1)} = 20 \pm 2$ ns

PRR = 200 kHz

FIGURE 6. Switching time test circuit and waveforms - Continued.

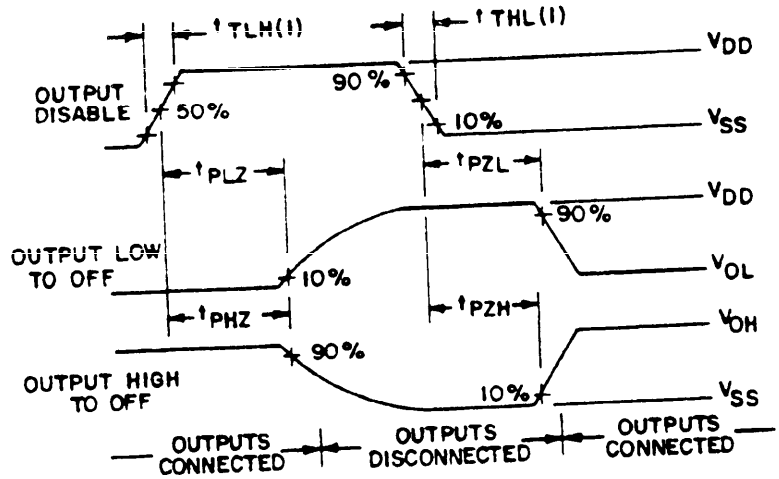
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Device type 03



Test circuit for tri-state
(Output disable to output)

R_L = 10 kΩ ±5%
C_L = 50 pF ±5%
(including wiring and probe capacitance)
SWI in position V_{DD} for t_{PLZ} and t_{PZL}
SWI in position V_{SS} for t_{PHZ} and t_{PZH}



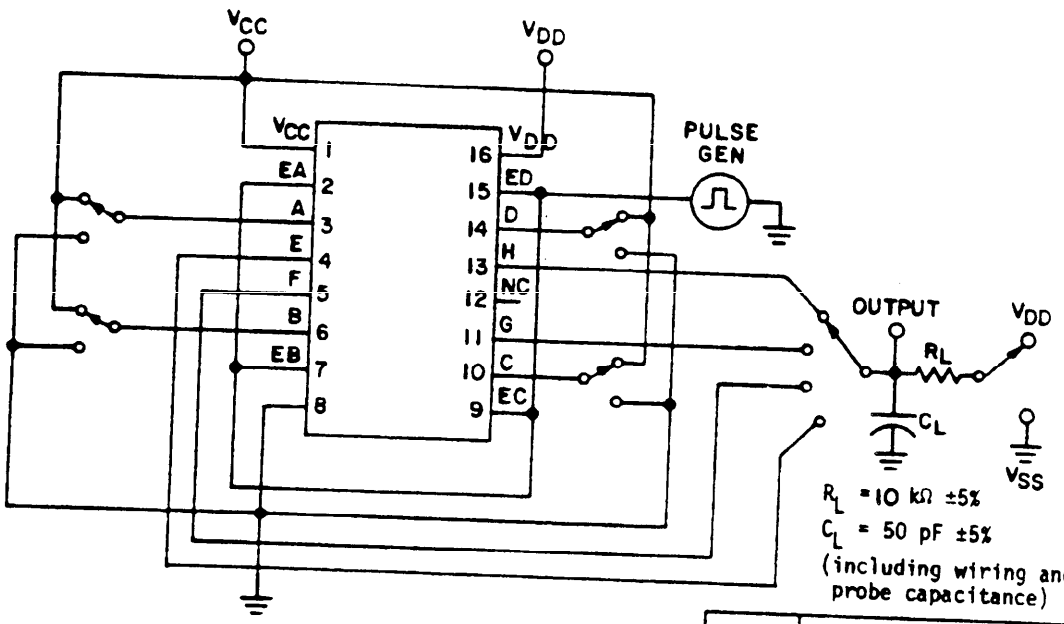
Dynamic test waveforms

Input pulse
$V_{GEN} = V_{DD}$ ±1.0%
t_{PH} = 1.0 ±0.1 µs
$t_{THL(1)} = t_{TLH(1)}$ = 20 ±2 ns
PRR = 200 kHz

FIGURE 6. Switching time test circuit and waveforms - Continued.

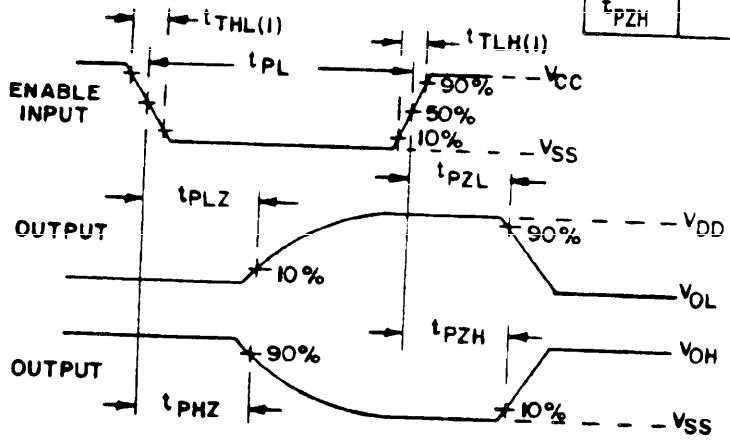
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Device type 04



Test	Test voltage	
	A, B, C, D	E, F, G, H
t_{PHZ}	V_{CC}	V_{SS}
t_{PLZ}	V_{SS}	V_{DD}
t_{PZL}	V_{SS}	V_{DD}
t_{PZH}	V_{CC}	V_{SS}

Test circuit for tri-state
(Output enable to output)



Dynamic test waveforms

$V_{GEN} = V_{CC} \pm 1.0\%$
$t_{PL} = 1.0 \pm 0.1$ µs
$t_{THL(1)} = t_{TLH(1)} = 20 \pm 2$ ns
PRR = 200 kHz

FIGURE 6. Switching time test circuit and waveforms - Continued.

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TABLE III. Group A inspection for device type 01.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	Terminal conditions 1/														Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	A	G	B	H	C	I	V_{SS}	J	D	K	E	L	F	V_{DD}								
		Test No.																Min	Max	Min	Max	Min	Max	
$V_{IC(pos)}$		1	1 mA													GND	A		1.5					Vdc
		2			1 mA												B							
		3					1 mA										C							
		4									1 mA						D							
		5											1 mA				E							
		6													1 mA		F							
$V_{IC(neg)}$		7	-1 mA						GND								A		-6.0					
		8			-1 mA												B							
		9					-1 mA										C							
		10									-1 mA						D							
		11											-1 mA				E							
		12													-1 mA		F							
I_{SS} 2/	3005	13	18.0 V		18.0 V		18.0 V				18.0 V		18.0 V		18.0 V	18.0 V	V_{SS}		-25		-750			nA
I_{SS} 2/	3005	14	GND		GND		GND				GND		GND		GND	18.0 V	V_{SS}		-25		-750			nA
V_{OL1}	3007	15	15.0 V		GND		GND									15.0 V	G		0.05		0.05		0.05	Vdc
		16	GND		15.0 V		GND										H							
		17			GND		15.0 V										I							
		18					GND				15.0 V						J							
		19					GND				GND		15.0 V				K							
		20					GND				GND		GND		15.0 V		L							
V_{OH1}	3006	21	GND		15.0 V		15.0 V				15.0 V		15.0 V		15.0 V		G	14.95		14.95		14.95		Vdc
		22	15.0 V		GND		15.0 V				15.0 V						H							
		23			15.0 V		GND				15.0 V						I							
		24					15.0 V				GND						J							
		25					15.0 V				15.0 V		GND				K							
		26					15.0 V				15.0 V		15.0 V		GND		L							

See notes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	Terminal conditions 1/														Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 $T_A = 25^\circ C$		Subgroup 2 $T_A = 125^\circ C$		Subgroup 3 $T_A = -55^\circ C$		
			A	G	B	H	C	I	V_{SS}	J	D	K	E	L	F	V_{DD}								
		Test No.																Min	Max	Min	Max	Min	Max	
V_{IH2}		27	7.0 V		GND		GND		GND		GND		GND		GND	10.0 V	G		1.0		1.0		1.0	Vdc
		28	GND		7.0 V		GND				GND						H							
		29			GND		7.0 V				GND						I							
		30					GND				7.0 V						J							
		31									GND		7.0 V				K							
		32											GND		7.0 V		L							
V_{IL2}		33	3.0 V												GND		G	9.0		9.0		9.0		
		34	GND		3.0 V												H							
		35			GND		3.0 V										I							
		36					GND				3.0 V						J							
		37					GND				GND		3.0 V				K							
		38					GND						GND		3.0 V		L							
V_{IH1}		39	3.5 V		GND		GND								GND	5.0 V	G		0.5		0.5		0.5	
		40	GND		3.5 V		GND										H							
		41			GND		3.5 V										I							
		42					GND				3.5 V						J							
		43									GND		3.5 V				K							
		44											GND		3.5 V		L							
V_{IL1}		45	1.5 V												GND		G	4.5		4.5		4.5		
		46	GND		1.5 V												H							
		47			GND		1.5 V										I							
		48					GND				1.5 V						J							
		49									GND		1.5 V				K							
		50									GND		GND		1.5 V		L							

See notes at end of table.

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TABLE III. Group A inspection for device type 01 -Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	Terminal conditions 1/														Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	A	G	B	H	C	J	V_{SS}	J	D	K	E	L	F	V_{DD}								
		Test No.																Min	Max	Min	Max	Min	Max	
V_{IH3}		51	11.0 V		GND		GND		GND		GND		GND		GND	15.0 V	G		1.5		1.5		1.5	Vdc
		52	GND		11.0 V		GND				GND						H							
		53			GND		11.0 V				GND						I							
		54					GND				11.0 V						J							
		55					GND				GND		11.0 V				K							
		56					GND				GND		GND		11.0 V		L							
V_{IL3}		57	4.0 V				GND				GND				GND		G	13.5		13.5		13.5		
		58	GND		4.0 V		GND				GND						H							
		59			GND		4.0 V				GND						I							
		60					GND				4.0 V						J							
		61					GND				GND		4.0 V				K							
		62					GND				GND		GND		4.0 V		L							
I_{IH1} 3/	3010	63	18.0 V		18.0 V		18.0 V				18.0 V		18.0 V		18.0 V	18 V	All inputs together		6					nA
I_{IH2}	3010	64	18.0 V		GND		GND				GND		GND		GND		A		1		45			
		65	GND		18.0 V		GND				GND						B							
		66			GND		18.0 V				GND						C							
		67					GND				18.0 V						D							
		68									GND		18.0 V				E							
		69									GND		GND		18.0 V		F							
I_{IL1} 3/	3009	70									GND		GND		GND		All inputs together		-6					
I_{IL2}	3009	71			18.0 V		18.0 V				18.0 V		18.0 V		18.0 V		A		-1		-45			
		72	18.0 V		GND		18.0 V				18.0 V						B		-1		-45			
		73	18.0 V		18.0 V		GND				18.0 V						C		-1		-45			

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	Terminal conditions 1/														Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	A	G	B	H	C	I	V_{SS}	J	D	K	E	L	F	V_{DD}								
		Test No.																Min	Max	Min	Max	Min	Max	
I_{IL2}	3009	74	18.0 V		18.0 V		18.0 V		GND		GND		18.0 V		18.0 V	18.0 V	D		-1		-45			nA
I_{IL2}	3009	75	18.0 V		18.0 V		18.0 V				18.0 V		GND		18.0 V		E		-1		-45			nA
I_{IL2}	3009	76	18.0 V		18.0 V		18.0 V				18.0 V		18.0 V		GND		F		-1		-45			nA
I_{OH1}		77	GND	4.6 V	GND		GND				GND		GND		GND	5.0 V	G	-0.51		-0.36		-0.64		mA
		78				4.6 V											H							
		79						4.6 V									I							
		80								4.6 V							J							
		81										4.6 V					K							
		82												4.6 V			L							
I_{OH2}		83		13.5 V												15.0 V	G	-3.4		-2.4		-4.2		
		84				13.5 V											H							
		85						13.5 V									I							
		86								13.5 V							J							
		87										13.5 V					K							
		88												13.5 V			L							
I_{OL1}		89	5.0 V	0.4 V	5.0 V		5.0 V				5.0 V		5.0 V		5.0 V	5.0 V	G	0.51		0.36		0.64		
		90				0.4 V											H							
		91						0.4 V									I							
		92								0.4 V							J							
		93										0.4 V					K							
		94												0.4 V			L							

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TABLE III Group A inspection for device type 01 -Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	Terminal conditions 1/													Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	A	G	B	H	C	I	V_{SS}	J	D	K	E	L	F	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																						
I_{OL2}		95	15.0 V	1.5 V	15.0 V	1.5 V	15.0 V	1.5 V	GND	1.5 V	15.0 V	1.5 V	15.0 V	1.5 V	15.0 V	15.0 V	G	3.4		2.4		4.2		mA
		96															H							
		97															I							
		98															J							
		99															K							
		100															L							
																		Subgroup 4 T_A = 25°C						
																		Min	Max					
C_I	3012	101	4/		4/		4/		GND	4/		4/		4/		GND	A		7.5					pF
		102															B							
		103															C							
		104															D							
		105															E							
		106															F							
																		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
																		Min	Max	Min	Max	Min	Max	
t_{PHL}	3003 Fig. 6	107	IN	OUT	GND	OUT	GND	OUT	GND	OUT	GND	OUT	GND	OUT	GND	5.0 V	A to G	5	110	8	155	5	110	ns
		108	GND		IN		GND				GND						B to H							
		109			GND		IN				GND						C to I							
		110					GND				IN						D to J							
		111									GND		IN				E to K							
		112											GND		IN		F to L							

See notes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	Terminal conditions 1/														Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Test No.	A	G	B	H	C	I	V_{SS}	J	D	K	E	L	F	V_{DD}		Min	Max	Min	Max	Min	Max	
t_{PLH}	3003 Fig. 6	113	IN	OUT	GND		GND		GND		GND		GND		GND	5.0 V	A to G	5	110	8	155	5	110	ns
		114	GND		IN	OUT	GND				GND						B to H							
		115			GND		IN	OUT			GND						C to I							
		116					GND			OUT	IN						D to J							
		117									GND	OUT	IN				E to K							
		118											GND	OUT	IN		F to L							
t_{THL}	3004 Fig. 6	119	IN	OUT											GND		G	10	200	14	280	10	200	
		120	GND		IN	OUT											H							
		121			GND		IN	OUT									I							
		122					GND			OUT	IN						J							
		123					GND				GND	OUT	IN				K							
		124					GND				GND		GND	OUT	IN		L							
t_{TLH}		125	IN	OUT	GND		GND				GND				GND		G							
		126	GND		IN	OUT	GND				GND						H							
		127			GND		IN	OUT			GND						I							
		128					GND			OUT	IN						J							
		129					GND				GND	OUT	IN				K							
		130					GND				GND		GND	OUT	IN		L							

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TABLE III Group A inspection for device type 01 -Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	Terminal conditions 1/														Measured terminal	Test limits		Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 12 T_A = 25°C		
		Symbol	A	G	B	H	C	I	V_{SS}	J	D	K	E	L	F	V_{DD}				
		Test No.																Min	Max	
t_{PHL}	3003 Fig. 6	131	IN	OUT	GND		GND		GND		GND		GND		GND	10.0 V	A to G	3	60	ns
		132	GND		IN	OUT	GND										B to H			
		133			GND		IN	OUT									C to I			
		134					GND			OUT	IN						D to J			
		135					GND				GND	OUT	IN				E to K			
		136					GND				GND		GND	OUT	IN		F to L			
t_{PLH}		137	IN	OUT	GND		GND				GND				GND		A to G			
		138	GND		IN	OUT	GND										B to H			
		139			GND		IN	OUT									C to I			
		140					GND			OUT	IN						D to J			
		141					GND				GND	OUT	IN				E to K			
		142					GND				GND		GND	OUT	IN		F to L			
t_{THL}	3004 Fig. 6	143	IN	OUT	GND		GND				GND				GND		G	5	100	
		144	GND		IN	OUT	GND				GND						H			
		145			GND		IN	OUT			GND						I			
		146					GND			OUT	IN						J			
		147									GND	OUT	IN				K			
		148											GND	OUT	IN		L			
t_{TLH}		149	IN	OUT	GND										GND		G			
		150	GND		IN	OUT											H			
		151			GND		IN	OUT									I			
		152					GND			OUT	IN						J			
		153					GND				GND	OUT	IN				K			
		154					GND				GND		GND	OUT	IN		L			

See notes at end of table.

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TABLE III. Group A inspection for device type 02.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	Terminal connections 1/														Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	NC	NC	A	B	C	NC	V_{SS}	NC	F	E	D	NC	NC	V_{DD}								
		Test No.																Min	Max	Min	Max	Min	Max	
$V_{IC(pos)}$		1			1 mA											GND	A		1.5					Vdc
		2				1 mA											B							
		3											1 mA				D							
		4										1 mA					E							
$V_{IC(neg)}$		5			-1 mA				GND								A		-6.0					
		6				-1 mA											B							
		7											-1 mA				D							
		8										-1 mA					E							
I_{SS} 2/	3005	9			GND	GND						GND	GND			18.0 V	V_{SS}		-75		-750			nA
		10			GND	18.0 V						18.0 V	GND											
		11			18.0 V	GND						GND	18.0 V											
		12			18.0 V	18.0 V						18.0 V	18.0 V											
		13			GND	GND						GND	GND											
V_{OL1}	3007	14			15.0 V	15.0 V						GND	GND			15.0 V	C		0.05		0.05		0.05	Vdc
V_{OL1}	3007	15			GND	GND						15.0 V	15.0 V			15.0 V	F		0.05		0.05		0.05	
V_{IH1}		16			3.5 V	GND	5/				5/	GND	GND			5.0 V	C	4.5		4.5		4.5		
		17			GND	3.5 V						GND	GND				C							
		18			GND	GND						GND	3.5 V				F							
		19			GND	GND						3.5 V	GND				F							
V_{IL1}		20			1.5 V	3.5 V						3.5 V	3.5 V				C							
		21			3.5 V	1.5 V						3.5 V	3.5 V				C							
		22			3.5 V	3.5 V						3.5 V	1.5 V				F							
		23			3.5 V	3.5 V						1.5 V	3.5 V				F							

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TABLE III. Group A inspection for device type 02 -Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	Terminal conditions 1/														Measured terminal	Test limits						Units
																		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	1	2	3	4	5	6	7	8	9	10	11	12	13	14								
		Test No.	NC	NC	A	B	C	NC	V_{SS}	NC	F	E	D	NC	NC	V_{DD}		Min	Max	Min	Max	Min	Max	
V_{IH2}		24			7.0 V	GND	5/		GND		5/	GND	GND			10.0 V	C	9.0		9.0		9.0		Vdc
		25			GND	7.0 V						GND	GND				C							
		26			GND	GND						GND	7.0 V				F							
		27			GND	GND						7.0 V	GND				F							
V_{IL2}		28			3.0 V	7.0 V						7.0 V	7.0 V				C							
		29			7.0 V	3.0 V							7.0 V				C							
		30			7.0 V	7.0 V							3.0 V				F							
		31			7.0 V	7.0 V						3.0 V	7.0 V				F							
V_{IH3}		32			11.0 V	GND						GND	GND			15.0 V	C	13.5		13.5		13.5		
		33			GND	11.0 V						GND	GND				C							
		34			GND	GND						GND	11.0 V				F							
		35			GND	GND						11.0 V	GND				F							
V_{IL3}		36			4.0 V	11.0 V						11.0 V	11.0 V				C							
		37			11.0 V	4.0 V							11.0 V				C							
		38			11.0 V	11.0 V							4.0 V				F							
		39			11.0 V	11.0 V						4.0 V	11.0 V				F							
I_{IH1} 3/	3010	40			18.0 V	18.0 V						18.0 V	18.0 V			18.0 V	All inputs together		4					nA
I_{IH2}	3010	41			18.0 V	GND						GND	GND				A		1		45			
		42			GND	18.0 V						GND	GND				B							
		43			GND	GND						GND	18.0 V				D							
		44			GND	GND						18.0 V	GND				E							
I_{IL1} 3/	3009	45			GND	GND						GND	GND				All inputs together		-4					

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TABLE III. Group A inspection for device type 03 -Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	Terminal conditions 1/														Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	NC	NC	A	B	C	NC	V_{SS}	NC	F	E	D	NC	NC	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																						
I_{IL2}	3009	46			GND	18.0 V			GND			18.0 V	18.0 V			18.0 V	A		-1		-45			nA
		47			18.0 V	GND						18.0 V	18.0 V				B							
		48			18.0 V	18.0 V						18.0 V	GND				D							
		49			18.0 V	18.0 V						GND	18.0 V				E							
I_{OL1}		50			5.0 V	5.0 V	0.4 V					5.0 V	5.0 V			5.0 V	C	16		21		12		mA
I_{OL1}		51			5.0 V	5.0 V					0.4 V	5.0 V	5.0 V			5.0 V	F	16		21		12		
I_{OL2}		52			15.0 V	15.0 V	1.5 V					15.0 V	15.0 V			15.0 V	C	110		140		80		
I_{OL2}		53			15.0 V	15.0 V					1.5 V	15.0 V	15.0 V			15.0 V	F	110		140		80		
																		Subgroup 4 T_A = 25°C						
																		Min	Max					
C_I	3012	54			4/				GND							GND	A		7.5					pF
		55				4/											B							
		56											4/				D							
		57										4/					E							
																		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
																		Min	Max	Min	Max	Min	Max	
t_{PHL}	3003 Fig. 6 6/	58			IN	5.0 V	OUT		GND			5.0 V	5.0 V			5.0 V	A to C	10	200	14	280	10	200	ns
		59			5.0 V	IN	OUT					5.0 V	5.0 V				B to C							
		60			5.0 V	5.0 V					OUT	5.0 V	IN				D to F							
		61			5.0 V	5.0 V					OUT	IN	5.0 V				E to F							
t_{PLH}		62			IN	5.0 V	OUT					5.0 V	5.0 V				A to C							
		63			5.0 V	IN	OUT					5.0 V	5.0 V				B to C							
		64			5.0 V	5.0 V					OUT	5.0 V	IN				D to F							
		65			5.0 V	5.0 V					OUT	IN	5.0 V				E to F							

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	Terminal conditions 1/														Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Symbol	NC	NC	A	B	C	NC	V_{SS}	NC	F	E	D	NC	NC	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																						
t_{THL}	3004 Fig. 6 6/	66			IN	5.0 V	OUT		GND			5.0 V	5.0 V			5.0 V	C	5	100	7	140	5	100	ns
		67			5.0 V	IN	OUT					5.0 V	5.0 V				C							
		68			5.0 V	5.0 V					OUT	5.0 V	IN				F							
		69			5.0 V	5.0 V					OUT	IN	5.0 V				F							
t_{TLH}		70			IN	5.0 V	OUT					5.0 V	5.0 V				C							
		71			5.0 V	IN	OUT					5.0 V	5.0 V				C							
		72			5.0 V	5.0 V					OUT	5.0 V	IN				F							
		73			5.0 V	5.0 V					OUT	IN	5.0 V				F							
																		Subgroup 12 T_A = 25°C						
																		Min	Max					
t_{PHL}	3003 Fig. 6 6/	74			IN	10.0 V	OUT		GND			10.0 V	10.0 V			10.0 V	A to C	5	90					ns
		75			10.0 V	IN	OUT					10.0 V	10.0 V				B to C							
		76			10.0 V	10.0 V					OUT	10.0 V	IN				D to F							
		77			10.0 V	10.0 V					OUT	IN	10.0 V				E to F							
t_{PLH}		78			IN	10.0 V	OUT					10.0 V	10.0 V				A to C	6	120					
		79			10.0 V	IN	OUT					10.0 V	10.0 V				B to C							
		80			10.0 V	10.0 V					OUT	10.0 V	IN				D to F							
		81			10.0 V	10.0 V					OUT	IN	10.0 V				E to F							
t_{THL}	3004 Fig. 6 6/	82			IN	10.0 V	OUT					10.0 V	10.0 V				C	2	40					
		83			OUT	IN	OUT					10.0 V	10.0 V				C							
		84			OUT	10.0 V					OUT	10.0 V	IN				F							
		85			OUT	10.0 V					OUT	IN	10.0 V				F							

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	Terminal conditions 1/														Measured terminal	Test limits			Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 12 $T_A = 25°C$			
		Symbol	NC	NC	A	B	C	NC	V_{SS}	NC	F	E	D	NC	NC	V_{DD}					
		Test No.																Min	Max		
t_{TLH}	3004 Fig. 6 2/	86			IN	10.0 V	OUT		GND			10.0 V	10.0 V			10.0 V	C	4	70		ns
		87			10.0 V	IN	OUT					10.0 V	10.0 V				C				
		88				10.0 V					OUT	10.0 V	IN				F				
		89				10.0 V					OUT	IN	10.0 V				F				

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TABLE III Group A inspection for device type 03.

Symbol	MIL-STD-883 method	Cases E, F, Z	Terminal conditions 1/															Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	D3	Q3	D1	Dis.	Q1	D2	Q2	V_{SS}	Q4	D4	Q5	I	D5	Q6	D6	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																								
$V_{IC(pos)}$		1				1 mA								1 mA				GND	Dis.		1.5					Vdc
		2																	I							
		3			1 mA														D_1							
		4						1 mA											D_2							
		5	1 mA																D_3							
		6										1 mA							D_4							
		7													1 mA				D_5							
		8															1 mA		D_6							
$V_{IC(neg)}$		9				-1 mA				GND									Dis.		-6.0					
		10												-1 mA					I							
		11			-1 mA														D_1							
		12						-1 mA											D_2							
		13	-1 mA																D_3							
		14										-1 mA							D_4							
		15													-1 mA				D_5							
		16															-1 mA		D_6							
I_{SS} 2/	3005	17	GND		GND	GND		GND				GND		GND	GND		GND	18.0 V	V_{SS}		-75		750			n/A
		18	18.0 V		18.0 V	GND		18.0 V				18.0 V		GND	18.0 V		18.0 V									
		19	18.0 V		18.0 V	GND		18.0 V				18.0 V		18.0 V	18.0 V		18.0 V									
		20	18.0 V		18.0 V	18.0 V		18.0 V				18.0 V		18.0 V	18.0 V		18.0 V									

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TABLE III. Group A inspection for device type 03 -Continued.

Symbol	MIL-STD-883 method	Cases E, F, Z	Terminal conditions 1/																Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	D3	Q3	D1	Dis.	Q1	D2	Q2	V_{SS}	Q4	D4	Q5	I	D5	Q6	D6	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																								
I_{OC11} 3/		21	18.0 V	18.0 V	18.0 V	18.0 V	18.0 V	18.0 V	18.0 V	GND	18.0 V	18.0 V	18.0 V	18.0 V	18.0 V	18.0 V	18.0 V	18.0 V	All output together		150					nA
I_{OC12}	8	22					18.0 V												Q_1		25		90		25	
		23							18.0 V										Q_2							
		24		18.0 V															Q_3							
		25									18.0 V								Q_4							
		26											18.0 V						Q_5							
		27														18.0 V			Q_6							
I_{OC21} 3/		28		GND			GND		GND		GND		GND			GND			All output together		-150					
I_{OC22}		29					GND												Q_1		-25		-90		-25	
		30							GND										Q_2							
		31		GND															Q_3							
		32									GND								Q_4							
		33											GND						Q_5							
		34														GND			Q_6							
V_{OH1}	3006	35	GND		GND	GND		GND				GND		GND	GND		GND	15.0 V	Q_1	14.95		14.95		14.95		Vdc
		36																	Q_2							
		37																	Q_3							
		38																	Q_4							
		39																	Q_5							
		40																	Q_6							

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TABLE III Group A inspection for device type 03 - Continued.

Symbol	MIL-STD-883 method	Cases E, F, Z	Terminal conditions 1/															Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	D3	Q3	D1	Dis.	Q1	D2	Q2	V_{SS}	Q4	D4	Q5	I	D5	Q6	D6	V_{DD}								
		Test No.																		Min	Max	Min	Max	Min	Max	
V_{OL1}	3007	41	15.0 V		15.0 V	GND		15.0 V		GND		15.0 V		GND	15.0 V		15.0 V	15.0 V	Q_1		0.05		0.05		0.05	Vdc
		42												15.0 V					Q_1							
		43												GND					Q_2							
		44												15.0 V					Q_2							
		45												GND					Q_3							
		46												15.0 V					Q_3							
		47												GND					Q_4							
		48												15.0 V					Q_4							
		49												GND					Q_5							
		50												15.0 V					Q_5							
		51												GND					Q_6							
		52												15.0 V					Q_6							
V_{IH1}		53	GND		3.5 V			GND				GND		GND	GND		GND	5.0 V	Q_1		0.5		0.5		0.5	Vdc
		54	GND		GND			3.5 V											Q_2							
		55	3.5 V					GND											Q_3							
		56	GND									3.5 V							Q_4							
		57										GND			3.5 V				Q_5							
		58													GND		3.5 V		Q_6							
		59												3.5 V			GND		Q_1							
		60																	Q_2							
		61																	Q_3							
		62																	Q_4							
		63																	Q_5							
		64																	Q_6							

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TABLE III. Group A inspection for device type 03 -Continued.

Symbol	MIL-STD-883 method	Cases E, F, Z	Terminal conditions 1/															Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	D3	Q3	D1	D4a.	Q1	D2	Q2	V_{SS}	Q4	D4	Q5	I	D5	Q6	D6	V_{DD}							
		Test No.																		Min	Max	Min	Max	Min	Max	
V_{IH2}		65	GND		7.0 V	GND		GND		GND		GND		GND	GND		GND	10.0 V	Q_1		1.0		1.0		1.0	Vdc
		66	GND		GND			7.0 V				GND							Q_2							
		67	7.0 V					GND				GND							Q_3							
		68	GND									7.0 V							Q_4							
		69										GND			7.0 V				Q_5							
		70													GND		7.0 V		Q_6							
		71												7.0 V			GND		Q_1							
		72																	Q_2							
		73																	Q_3							
		74																	Q_4							
		75																	Q_5							
		76																	Q_6							
V_{IH3}		77			11.0 V									GND				15.0 V	Q_1		1.5		1.5		1.5	
		78			GND			11.0 V											Q_2							
		79	11.0 V					GND											Q_3							
		80	GND									11.0 V							Q_4							
		81										GND			11.0 V				Q_5							
		82													GND		11.0 V		Q_6							
		83												11.0 V			GND		Q_1							
		84																	Q_2							
		85																	Q_3							
		86																	Q_4							
		87																	Q_5							
		88																	Q_6							

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TABLE III. Group A inspection for device type 03 -Continued.

Symbol	MIL-STD-883 method	Cases E, F, Z	Terminal conditions 1/																Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_A 25°C		Subgroup 2 T_A 125°C		Subgroup 3 T_A -55°C		
		Symbol	D3	Q3	D1	D1a	Q1	D2	Q2	V_{SS}	Q4	D4	Q5	I	D5	Q6	D6	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																								
V_{IL1}		89	GND		1.5 V	GND		GND		GND		GND		GND	GND		GND	5.0 V	Q_1	4.5		4.5		4.5		Vdc
		90	GND		GND			1.5 V											Q_2							
		91	1.5 V					GND											Q_3							
		92	GND									1.5 V							Q_4							
		93										GND			1.5 V				Q_5							
		94										GND			GND		1.5 V		Q_6							
		95	1.5 V		1.5 V			1.5 V				1.5 V		1.5 V	1.5 V				Q_1							
		96																	Q_2							
		97																	Q_3							
		98																	Q_4							
		99																	Q_5							
		100																	Q_6							
V_{IL2}		101	GND		3.0 V			GND				GND		GND	GND		GND	10.0 V	Q_1	9.0		9.0		9.0		
		102	GND		GND			3.0 V				GND							Q_2							
		103	3.0 V					GND				GND							Q_3							
		104	GND									3.0 V							Q_4							
		105	GND									GND			3.0 V				Q_5							
		106	GND									GND			GND		3.0 V		Q_6							
		107	3.0 V		3.0 V			3.0 V				3.0 V		3.0 V	3.0 V				Q_1							
		108																	Q_2							
		109																	Q_3							
		110																	Q_4							
		111																	Q_5							
		112																	Q_6							

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TABLE III. Group A inspection for device type 03 -Continued.

Symbol	MIL-STD-883 method	Cases E, F, Z	Terminal conditions 1/																Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	D3	Q3	D1	Dis.	Q1	D2	Q2	V_{SS}	Q4	D4	Q6	I	D5	Q6	D6	V_{DD}								
		Test No.																		Min	Max	Min	Max	Min	Max	
V_{IL3}		113	GND		4.0 V	GND		GND		GND		GND		GND	GND		GND	15.0 V	Q_1	13.5		13.5		13.5		Vdc
		114	GND		GND			4.0 V				GND							Q_2							
		115	4.0 V					GND				GND							Q_3							
		116	GND									4.0 V							Q_4							
		117	GND									GND			4.0 V				Q_5							
		118	GND									GND			GND		4.0 V		Q_6							
		119	4.0 V		4.0 V			4.0 V				4.0 V		4.0 V	4.0 V				Q_1							
		120																	Q_2							
		121																	Q_3							
		122																	Q_4							
		123																	Q_5							
		124																	Q_6							
I_{IH1} 3/	3010	125	18.0 V		18.0 V	18.0 V		18.0 V				18.0 V		18.0 V	18.0 V		18.0 V	18.0 V	All inputs together		8					nA
I_{IH2}		126	GND		GND	18.0 V		GND				GND		GND	GND		GND		Dis.		1		45			
		127			GND	GND		GND						18.0 V					I							
		128			18.0 V			GND						GND					D_1							
		129			GND			18.0 V											D_2							
		130	18.0 V					GND											D_3							
		131	GND									18.0 V							D_4							
		132										GND			18.0 V				D_5							
		133										GND			GND		18.0 V		D_6							
I_{IL1} 3/	3009	134										GND			GND		GND		All inputs together		-8					

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TABLE III. Group A inspection for device type 03 - Continued.

Symbol	MIL-STD-883 method	Cases E, F, Z	Terminal conditions 1/															Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_A 25°C		Subgroup 2 T_A 125°C		Subgroup 3 T_A -55°C		
		Symbol	D3	Q3	D1	Dis.	Q1	D2	Q2	V_{SS}	Q4	D4	Q5	I	D5	Q6	D6	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																								
I_{IL2}	3009	135	18.0 V		18.0 V	GND		18.0 V		GND		18.0 V		18.0 V	18.0 V		18.0V	18.0 V	Dis.		-1		-45			nA
		136			18.0 V	18.0 V		18.0 V						GND					I							
		137			GND			18.0 V						18.0 V					D_1							
		138			18.0 V			GND											D_2							
		139	GND					18.0 V											D_3							
		140	18.0 V									GND							D_4							
		141	18.0 V									18.0 V			GND				D_5							
		142	18.0 V									18.0 V			18.0 V		GND		D_6							
I_{OH1}		143	GND		GND	GND	4.6 V	GND				GND		GND	GND			5.0 V	Q_1	-0.51		-0.36		-0.64		mA
		144							4.6 V										Q_2							
		145		4.6 V															Q_3							
		146									4.6 V								Q_4							
		147											4.6 V						Q_5							
		148														4.6 V			Q_6							
I_{OL1}		149					0.4 V							5.0 V					Q_1	0.51		0.36		0.54		
		150							0.4 V										Q_2							
		151		0.4 V															Q_3							
		152									0.4 V								Q_4							
		153											0.4 V						Q_5							
		154														0.4 V			Q_6							
I_{OH2}		155					13.5 V							GND				15.0 V	Q_1	-3.4		-2.4		-4.2		
		156							13.5 V										Q_2							
		157		13.5 V															Q_3							
		158									13.5 V								Q_4							
		159											13.5 V						Q_5							
		160														13.5 V			Q_6							

See notes at end of table.

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TABLE III. Group A inspection for device type 03 -Continued.

Symbol	MIL-STD-883 method	Cases E, F, Z	Terminal conditions 1/															Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	D3	Q3	D1	Dis.	Q1	D2	Q2	V_{SS}	Q4	D4	Q5	I	D5	Q6	D6	V_{DD}								
		Test No.																		Min	Max	Min	Max	Min	Max	
I_{OL2}		161	GND		GND	GND	1.5 V	GND		GND		GND		15.0 V	GND		GND	15.0 V	O_1	3.4		2.4		4.2		mA
		162							1.5 V										O_2							
		163		1.5 V															O_3							
		164									1.5 V								O_4							
		165											1.5 V						O_5							
		166														1.5 V			O_6							
																				Subgroup 4 T_A = 25°C						
																				Min	Max					
C_I	3012	167				4/				GND								GND	Dis.		7.5					pF
		168												4/					I							
		169			4/														D_1							
		170						4/											D_2							
		171	4/																D_3							
		172										4/							D_4							
		173													4/				D_5							
		174															4/		D_6							

See notes at end of table.

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TABLE III. Group A inspection for device type 03 -Continued.

Symbol	MIL-STD-883 method	Cases E, F, Z	Terminal conditions 1/																Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Symbol	D3	Q3	D1	Dis.	Q1	D2	Q2	V_{SS}	Q4	D4	Q5	1	D5	Q6	D6	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																								
t_{PHL}	3003 Fig. 6	175	GND		GND	GND	OUT	GND		GND		GND		IN	GND		GND	5.0 V	I to Q_1	14	270	19	380	14	270	ns
		176							OUT										I to Q_2							
		177		OUT															I to Q_3							
		178									OUT								I to Q_4							
		179											OUT						I to Q_5							
		180														OUT			I to Q_6							
		181			IN		OUT							GND					D_1 to Q_1							
		182			GND			IN	OUT										D_2 to Q_2							
		183	IN	OUT				GND											D_3 to Q_3							
		184	GND								OUT	IN							D_4 to Q_4							
		185										GND	OUT		IN				D_5 to Q_5							
		186													GND	OUT	IN		D_6 to Q_6							
t_{PLH}	3003 Fig. 6	187					OUT							IN			GND		I to Q_1	19	380	27	535	19	380	
		188							OUT										I to Q_2							
		189		OUT															I to Q_3							
		190									OUT								I to Q_4							
		191											OUT						I to Q_5							
		192														OUT			I to Q_6							
		193			IN		OUT							GND					D_1 to Q_1							
		194			GND			IN	OUT										D_2 to Q_2							
		195	IN	OUT				GND											D_3 to Q_3							
		196	GND								OUT	IN							D_4 to Q_4							
		197										GND	OUT		IN				D_5 to Q_5							
		198										GND			GND	OUT	IN		D_6 to Q_6							

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TABLE III Group A inspection for device type 03 -Continued.

Symbol	MIL-STD-883 method	Cases E, F, Z	Terminal conditions 1/															Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 9 T_A = 25 °C		Subgroup 10 T_A = 125 °C		Subgroup 11 T_A -55 °C		
		Symbol	D3	Q3	D1	Dis.	Q1	D2	Q2	V_{SS}	Q4	D4	Q5	I	D5	Q6	D6	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																								
t_{THL}	3004 Fig. 6	199	GND		GND	GND	OUT	GND		GND		GND		IN	GND		GND	5.0 V	Q_1	6	120	9	170	6	120	ns
		200							OUT										Q_2							
		201		OUT															Q_3							
		202									OUT								Q_4							
		203											OUT						Q_5							
		204														OUT			Q_6							
		205			IN		OUT							GND					Q_1							
		206			GND			IN	OUT										Q_2							
		207	IN	OUT				GND											Q_3							
		208	GND								OUT	IN							Q_4							
		209										GND	OUT		IN				Q_5							
		210													GND	OUT	IN		Q_6							
t_{TLH}	3004 Fig. 6	211					OUT							IN			GND		Q_1	10	200	14	280	10	200	
		212							OUT										Q_2							
		213		OUT															Q_3							
		214									OUT								Q_4							
		215											OUT						Q_5							
		216														OUT			Q_6							
		217			IN		OUT							GND					Q_1							
		218			GND			IN	OUT										Q_2							
		219	IN	OUT				GND											Q_3							
		220	GND								OUT	IN							Q_4							
		221	GND									GND	OUT		IN				Q_5							
		222	GND									GND			GND	OUT	IN		Q_6							

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TABLE III. Group A inspection for device type 03 -Continued.

Symbol	MIL-STD-883 method	Cases E, F, Z	Terminal conditions 1															Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Symbol	D3	Q3	D1	Dis.	Q1	D2	Q2	V_{SS}	Q4	D4	Q5	I	D5	Q6	D6	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																								
t_{PHZ} 5/	Fig. 6	223	GND		GND	IN	OUT	GND		GND		GND		GND	GND		GND	5.0 V	Dis to Q_1	6	120	9	170	6	120	ns
		224							OUT										Dis to Q_2							
		225		OUT															Dis to Q_3							
		226									OUT								Dis to Q_4							
		227											OUT						Dis to Q_5							
		228														OUT			Dis to Q_6							
t_{PZH} 5/		229					OUT												Dis to Q_1	11	220	16	310	11	220	
		230							OUT										Dis to Q_2							
		231		OUT															Dis to Q_3							
		232									OUT								Dis to Q_4							
		233											OUT						Dis to Q_5							
		234														OUT			Dis to Q_6							
t_{PLZ} 5/		235	5.0 V		5.0 V		OUT	5.0 V				5.0 V			5.0 V		5.0 V		Dis to Q_1	13	250	18	350	13	250	
		236							OUT										Dis to Q_2							
		237		OUT															Dis to Q_3							
		238									OUT								Dis to Q_4							
		239											OUT						Dis to Q_5							
		240														OUT			Dis to Q_6							
t_{PZL} 5/		241					OUT												Dis to Q_1							
		242							OUT										Dis to Q_2							
		243		OUT															Dis to Q_3							
		244									OUT								Dis to Q_4							
		245											OUT						Dis to Q_5							
		246														OUT			Dis to Q_6							

See notes at end of table.

MIL-M-38510/127A

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TABLE III. Group A inspection for device type 03 -Continued.

Symbol	MIL-STD-883 method	Cases E, F, Z	Terminal conditions 1/															Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 12 $T_A = 25°C$					
		Symbol	D3	Q3	D1	Dis.	Q1	D2	Q2	V_{SS}	Q4	D4	Q5	I	D5	Q6	D6	V_{DD}		Min	Max				
		Test No.																							
t_{PHL}	3003 Fig. 6	247	GND		GND	GND	OUT	GND		GND		GND		IN	GND		GND	10.0 V	I to Q_1	6	120				ns
		248							OUT										I to Q_2						
		249		OUT															I to Q_3						
		250									OUT								I to Q_4						
		251											OUT						I to Q_5						
		252														OUT			I to Q_6						
		253			IN		OUT							GND					D_1 to Q_1						
		254			GND			IN	OUT										D_2 to Q_2						
		255	IN	OUT				GND											D_3 to Q_3						
		256	GND								OUT	IN							D_4 to Q_4						
		257										GND	OUT		IN				D_5 to Q_5						
		258													GND	OUT	IN		D_6 to Q_6						
t_{PLH}	3003 Fig. 6	259					OUT							IN			GND		I to Q_1	9	180				
		260							OUT										I to Q_2						
		261		OUT															I to Q_3						
		262									OUT								I to Q_4						
		263											OUT						I to Q_5						
		264														OUT			I to Q_6						
		265			IN		OUT							GND					D_1 to Q_1						
		266			GND			IN	OUT										D_2 to Q_2						
		267	IN	OUT				GND											D_3 to Q_3						
		268	GND								OUT	IN							D_4 to Q_4						
		269	GND									GND	OUT		IN				D_5 to Q_5						
		270	GND									GND			GND	OUT	IN		D_6 to Q_6						

See notes at end of table.

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TABLE III. Group A inspection for device type 03 - Continued.

Symbol	MIL-STD-883 method	Cases E, F, Z	Terminal conditions 1/															Measured terminal	Test limits		Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Subgroup 12 T_A = 25°C		
		Symbol	D3	Q3	D1	Dis.	Q1	D2	Q2	V_{SS}	Q4	D4	Q5	I	D5	Q6	D6	V_{DD}			
		Test No.																	Min	Max	
t_{THL}	3004 Fig. 6	271	GND		GND	GND	OUT	GND		GND		GND		IN	GND		GND	10.0 V	Q_1	3	60	ns
		272							OUT										Q_2			
		273		OUT															Q_3			
		274									OUT								Q_4			
		275											OUT						Q_5			
		276														OUT			Q_6			
		277			IN		OUT							GND					Q_1			
		278			GND			IN	OUT										Q_2			
		279	IN	OUT				GND											Q_3			
		280	GND								OUT	IN							Q_4			
		281										GND	OUT		IN				Q_5			
		282													GND	OUT	IN		Q_6			
t_{TLH}	3004 Fig. 6	283					OUT							IN			GND		Q_1	5	100	
		284							OUT										Q_2			
		285		OUT															Q_3			
		286									OUT								Q_4			
		287											OUT						Q_5			
		288														OUT			Q_6			
		289			IN		OUT							GND					Q_1			
		290			GND			IN	OUT										Q_2			
		291	IN	OUT				GND											Q_3			
		292	GND								OUT	IN							Q_4			
		293	GND									GND	OUT		IN				Q_5			
		294	GND									GND			GND	OUT	IN		Q_6			

See notes at end of table.

MIL-M-38510/174A

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TABLE III. Group A inspection for device type 03 -Continued.

Symbol	MIL-STD-883 method	Cases E, F, Z	Terminal conditions 1/																Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 12 T_A = 25°C						
		Symbol	D3	Q3	D1	Dis.	Q1	D2	Q2	V_{SS}	Q4	D4	Q5	I	D5	Q6	D6	V_{DD}								
		Test No.																		Min	Max					
t_{PHZ} 5/	Fig. 6	295	GND		GND	IN	OUT	GND		GND		GND		GND	GND		GND	10.0 V	Dis to Q_1	4	80					ns
		296							OUT										Dis to Q_2							
		297		OUT															Dis to Q_3							
		298									OUT								Dis to Q_4							
		299											OUT						Dis to Q_5							
		300														OUT			Dis to Q_6							
t_{PZH} 5/		301					OUT												Dis to Q_1	5	100					
		302							OUT										Dis to Q_2							
		303		OUT															Dis to Q_3							
		304									OUT								Dis to Q_4							
		305											OUT						Dis to Q_5							
		306														OUT			Dis to Q_6							
t_{PLZ} 5/		307	10.0 V		10.0 V		OUT	10.0 V				10.0 V			10.0 V		10.0 V		Dis to Q_1	7	130					
		308							OUT										Dis to Q_2							
		309		OUT															Dis to Q_3							
		310									OUT								Dis to Q_4							
		311											OUT						Dis to Q_5							
		312														OUT			Dis to Q_6							
t_{PZL} 5/		313					OUT												Dis to Q_1	6	110					
		314							OUT										Dis to Q_2							
		315		OUT															Dis to Q_3							
		316									OUT								Dis to Q_4							
		317											OUT						Dis to Q_5							
		318														OUT			Dis to Q_6							

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TABLE III. Group A inspection for device type 04.

Symbol	MIL-STD-883 method	Cases E, F, Z	Terminal conditions 1/																Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	V_{CC}	EA	IA	E	F	IB	EB	V_{SS}	EC	IC	G	NC	H	ID	ED	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																								
$V_{IC(neg)}$		1	GND		-1 mA					GND									1A		-6.0					Vdc
		2		-1 mA															EA							
		3						-1 mA											IB							
		4							-1 mA										EB							
		5									-1 mA								EC							
		6										-1 mA							IC							
		7														-1 mA			ID							
		8															-1 mA		ED							
I_{SS}	3005	9	18.0 V	18.0 V	GND			GND	18.0 V		18.0 V	GND				GND	18.0 V	18.0 V	V_{SS}		-75		-750			nA
		10	18.0 V	18.0 V	18.0 V			18.0 V	18.0 V		18.0 V	18.0 V				18.0 V	18.0 V									
		11	18.0 V	GND	18.0 V			18.0 V	GND		GND	18.0 V				18.0 V	GND									
V_{OH1}	3006	12	5.0 V	15.0 V	15.0 V			15.0 V	15.0 V		15.0 V	15.0 V				15.0 V	15.0 V	15.0 V	E	14.95		14.95		14.95		Vdc
		13																	F							
		14																	G							
		15																	H							
V_{OL1}	3007	16			GND			GND				GND				GND			E		0.05		0.05		0.05	
		17																	F							
		18																	G							
		19																	H							
I_{OC11} 3/		20		GND		18.0 V	18.0 V		GND		GND		18.0 V		18.0 V		GND	18.0 V	All output together		100					nA
I_{OC12}		21				18.0 V													E		25		90		25	
		22					18.0 V												F							
		23											18.0 V						G							
		24													18.0 V				H							

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TABLE III. Group A inspection for device type 04 -Continued.

Symbol	MIL-STD-883 method	Cases E, F, Z	Terminal conditions 1/																Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	V_{CC}	EA	I A	E	F	IB	EB	V_{SS}	EC	IC	C	NC	H	ID	ED	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																								
I_{OC2} 2/		25	5.0 V	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND		GND	GND	GND	18.0 V	All output together		-100					nA
I_{OC22}		26				GND													E		-25		-90		-25	
		27					GND												F							
		28											GND						G							
		29													GND				H							
V_{IH}		30		3.5 V	5.0 V			5.0 V	5.0 V		5.0 V	5.0 V				5.0 V	5.0 V	10.0 V	E	9.0		9.0		9.0		Vdc
		31		5.0 V					3.5 V										F							
		32							5.0 V		3.5 V								G							
		33									5.0 V						3.5 V		H							
		34			3.5 V												5.0 V		E							
		35			5.0 V			3.5 V											F							
		36			5.0 V			5.0 V				3.5 V							G							
		37			5.0 V			5.0 V				5.0 V				3.5 V			H							
		38		3.5 V	GND			GND				GND				GND		15.0 V	E		1.5		1.5		1.5	
		39		5.0 V					3.5 V										F							
		40							5.0 V		3.5 V								G							
		41									5.0 V						3.5 V		H							
		42			3.5 V												5.0 V		E	13.5		13.5		13.5		
		43			GND			3.5 V											F							
		44			GND			GND				3.5 V							G							
		45			GND			GND				GND				3.5 V			H							
		46	10.0 V	7.0 V	10.0 V			10.0 V	10.0 V		10.0 V	10.0 V				10.0 V	10.0 V		E							
		47	10.0 V	10.0 V	10.0 V			10.0 V	7.0 V		10.0 V					10.0 V	10.0 V		F							
		48	10.0 V	10.0 V	10.0 V			10.0 V	10.0 V		7.0 V					10.0 V	10.0 V		G							

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TABLE III. Group A inspection for device type 04 - Continued.

Symbol	MIL-STD-883 method	Cases E, F, Z	Terminal conditions 1/																Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	V_{CC}	EA	IA	E	F	IB	EB	V_{SS}	EC	IC	G	NC	H	ID	ED	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																								
I_{OL2}		95	5.0 V	15.0 V	GND		1.5 V	GND	15.0 V	GND	15.0 V	GND	1.5 V		1.5 V	GND	15.0 V	15.0 V	F	3.4		2.4		4.2		mA
		96											1.5 V						G							
		97																	G							
		98													1.5 V				H							
		99													1.5 V				H							
																				Subgroup 4 T_A = 25°C						
																				Min	Max					
C_I	3012	100	GND	4/	4/			4/	4/	GND	4/	4/				4/	4/	GND	EA		7.5					pF
		101																	IA							
		102																	IB							
		103																	EB							
		104																	EC							
		105																	IC							
		106																	ID							
		107																	ED							
																				Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
																				Min	Max	Min	Max	Min	Max	
t_{PHL}	3003 Fig. 6	108	5.0 V	5.0 V	IN	OUT	OUT	GND	5.0 V	GND	5.0 V	GND	OUT		OUT	GND	5.0 V	10 V	IA to E	30	600	42	840	30	600	ns
		109			GND			IN				GND				GND			IB to F							
		110			GND			GND				IN				GND			IC to G							
		111			GND			GND				GND				IN			ID to H							

See notes at end of table.

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TABLE III. Group A inspection for device type 04 -Continued.

Symbol	MIL-STD-883 method	Cases E, F, Z	Terminal conditions 1/																Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Symbol	V_{CC}	EA	IA	E	F	IB	EB	V_{SS}	EC	IC	G	NC	H	ID	ED	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																								
t_{PLH}	3003 Fig. 6	112	5.0 V	5.0 V	IN	OUT		GND	5.0 V	GND	5.0 V	GND				GND	5.0 V	10 V	IA to E	13	260	19	365	13	260	ns
		113			GND		OUT	IN				GND				GND			IB to F							
		114			GND			GND				IN	OUT			GND			IC to G							
		115			GND			GND				GND			OUT	IN			ID to H							
t_{TLH}	3004 Fig. 6	116			IN	OUT		GND				GND				GND			IA to E	5	100	7	140	5	100	
		117			GND		OUT	IN				GND				GND			IB to F							
		118			GND			GND				IN	OUT			GND			IC to G							
		119			GND			GND				GND			OUT	IN			ID to H							
t_{THL}		120			IN	OUT		GND				GND				GND			IA to E							
		121			GND		OUT	IN				GND				GND			IB to F							
		122						GND				IN	OUT			GND			IC to G							
		123										GND			OUT	IN			ID to H							
t_{PLZ} 5/	Fig. 6	124		IN		OUT										GND			EA to E	37	740	52	1040	37	740	
		125		GND			OUT		IN										EB to F							
		126		GND					GND		IN		OUT						EC to G							
		127		GND					GND		GND				OUT		IN		ED to H							
t_{PZL} 5/		128		IN		OUT			GND		GND						GND		EA to E	10	200	14	280	10	200	
		129		GND			OUT		IN		GND						GND		EB to F							
		130		GND					GND		IN		OUT				GND		EC to G							
		131		GND					GND		GND				OUT		IN		ED to H							
t_{PZH} 5/		132		IN	5.0 V	OUT		5.0 V	GND		GND	5.0 V				5.0 V	GND		EA to E	32	640	45	895	32	640	
		133		GND			OUT		IN		GND						GND		EB to F							
		134		GND					GND		IN		OUT				GND		EC to G							
		135		GND					GND		GND				OUT		IN		ED to H							

See notes at end of table.

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TABLE III. Group A inspection for device type 04 -Continued.

Symbol	MIL-STD-883 method	Cases E, F, Z	Terminal conditions 1/															Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 9 T_A=25°C		Subgroup 10 T_A=125°C		Subgroup 11 T_A=-55°C		
		Symbol	V_{CC}	EA	IA	E	F	IB	EB	V_{SS}	EC	IC	G	NC	H	ID	ED	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																								
t_{PHZ} 5/	Fig. 6	136	5.0 V	IN	5.0 V	OUT		5.0 V	GND	GND	GND	5.0 V				5.0 V	GND	10 V	EA to E	6	120	9	170	6	120	ns
		137		GND			OUT		IN		GND		OUT				GND		EB to F							
		138		GND					GND		IN						GND		EC to G							
		139		GND					GND		GND				OUT		IN		ED to H							

NOTES:

1/ Pins not designated may be "HIGH" level logic, "LOW" level logic or open. Exceptions are as follows: $V_{IC(pos)}$ tests, the V_{SS} terminals shall be open; $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open; I_{SS} tests, the output terminals shall be open.

2/ The I_{SS} measurements shall be performed in sequence.

3/ The device manufacturer may, at his option, measure I_{IL}, I_{IH}, I_{OCH}, and I_{OC2} at 25°C for each individual input or measure all inputs together.

4/ See 4.4.1c.

5/ Measured with external pull-up resistor of 10 kΩ ±5% connected between the output and V_{DD} for t_{PLZ} and t_{PZL} ; and V_{SS} for t_{PHZ} and t_{PZH}.

6/ For device type 02 during the transition and propagation measurements, the output shall use an external pull up resistor of 120Ω connected between output and V_{DD}.

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b. A special subgroup shall be added using an LTPD of 15 for classes S and B, and measured only for initial qualification and after process or design changes. This subgroup shall consist of a high voltage test of the input protection circuits, V_{ZAP} (see 4.5.3).

c. End-point electrical parameters shall be as specified in table II herein and shall consist only of those subgroups specified in table IIa of test method 5005 of MIL-STD-883, and table II herein also. Delta limits shall apply only to subgroup 5 of group B inspections and shall consist of tests specified in table IV herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection and shall consist of tests specified in table IV herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D and as specified in 4.5.2 and as shown on figure 5, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration, 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

c. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B devices, and shall consist of the tests, conditions, and limits as specified for subgroups 10 and 11 of group A.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.4.5 Group E inspection. Group E inspection is required only for device types intended to be marked as radiation hardened (see 3.6.1). When group E testing is performed it shall be in accordance with table V of method 5005 of MIL-STD-883 and 4.5.5 herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit V_{SS} terminal, unless otherwise specified. Currents given are conventional and positive when flowing into the referenced terminal.

4.5.2 Burn-in and life test cool down procedures. When these tests are completed and prior to removal of bias voltages, the devices under test (DUT) shall be cooled to a temperature of 25° ±3°C; then, electrical parameter end-point measurements shall be performed.

TABLE IV. Delta limits at 025°C.

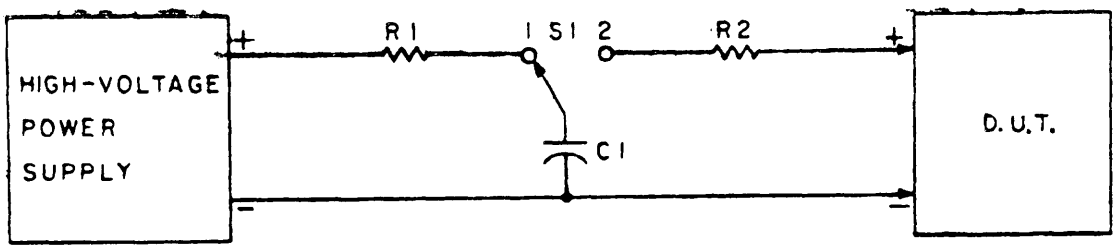
Parameter 1/	Device types			
	01	02	03	04
I_{SS}	±10 nA	±10 nA	±10 nA	±10 nA
I_{OL1}	15%	15%	15%	15%
I_{OH1}	15%	15%	15%	15%

1/ Each of the above parameters shall be recorded before and after the required burn-in and life tests to determine delta's (Δ).

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4.5.3 High voltage (V_{ZAP}) test or input protection circuits. All input terminals (up to a maximum of 4) of the DUT shall be subjected to a voltage pulse, from a 100 pF source charged to 400 V. This destructive test shall be conducted as follows using the test circuit on figure 7:

a. Measure I_{IL} and I_{IH} at the inputs selected, as stated above, at 25°C. The test limit for each input tested shall be ±10 nA at the specified V_{DD}. Measure I_{SS} on the DUT at 25°C. The test limit for this measurement shall be increased a maximum of 20 percent of the specified I_{SS} table III limit at the specified V_{DD} 18.



V_{ZAP} = 400 V minimum charge on C1
1 MΩ ≤ R1 ≤ 50 MΩ
R2 = 1.5 kΩ
C1 = 100 pF
S1 = Hg-wetted "bounceless" relay

FIGURE 7. High voltage (V_{ZAP}) test circuit.

b. V_{ZAP} is applied to DUT in the following modes (see table V) by changing C1 to V_{ZAP} with S1 in position 1 and then switching to position 2.

TABLE V. Modes for high voltage test.

Mode	+ Terminal	- Terminal
1	V_{DD}	Input
2	Input	V_{SS}
3	Input	Associated Output

c. Within 24 hours repeat the I_{SS}, I_{IL}, and I_{IH} measurements on the same terminals as performed above. If a DUT exhibits leakage currents in excess of the specified limits after the V_{ZAP} test, it shall be classified as a failure.

4.5.4 Quiescent supply current (I_{SS}) test. When performing quiescent supply current measurements (I_{SS}), the meter shall be placed so that all currents flow through the meter.

4.5.5 Radiation hardness assurance (RHA) testing. The RHA testing shall be performed in accordance with test procedures and sampling specified in table V of method 5005 of MIL-STD-883 and herein:

a. Before irradiation, selected samples shall be assembled in qualified packages and pass the governing electrical parameters (group A subgroup 1 at 25°C) and also be subjected to the threshold-voltage test in table VIII in order to calculate the delta threshold (ΔV_T) after irradiation.

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b. The devices shall be subjected to a total radiation dose as specified in MIL-M-38510 for the radiation hardness assurance (RHA) level being tested, and meet the end point electrical parameters as defined in table VI at 25°C, after exposure. The start and completion of the end point electrical parameter measurements shall not exceed 2 hours following irradiation.

c. Threshold-voltage test circuit conditions shall be as specified in table VIII and figure 8. In situ and remote testing, the test shall be performed with the devices biased in accordance with table VII and bias may be interrupted for up to 1 minute to remove devices to the remote bias fixture.

d. After irradiation, the devices shall pass the truth table test as specified in subgroup 7 in table III or if subgroup 7 is not required, then an equivalent truth table test shall be performed.

TABLE VI. Radiation hardening end-point electrical parameters at 25°C.

Parameter	All device types	V_{DD}
V_{TN}	0.3 V min	10 V
V_{TP}	2.8 V max	10 V
ΔV_T	1.4 V	10 V
I_{SS}	100 x max limit	18 V
t_{PLH}	1.35 x max limit	5 V
t_{PHL}	1.35 x max limit	5 V

TABLE VII. Bias during exposure to radiation.

Device type	Pin connections		
	V_{DD} = 10 V dc (through a 30 k to 60 k resistor)	V_{SS} = GND	V_{DD} = 10 V dc
01	1, 3, 5, 8, 9, 11, 13	7	14
02	3, 4, 5, 9, 10, 11	7	14
03	1, 3, 4, 6, 10, 12, 13, 15	8	16
04	2, 3, 6, 7, 9, 10, 14, 15	8	1, 16

Pins not designated are open or connected to 10 V dc through a 30 kΩ to 60 resistor.

4.6 Data reporting. When specified in the purchase order or contract, a copy of the following data, as applicable, shall be supplied.

a. Attributes data for all screening tests (see 4.2) and variables data for all static burn-in, dynamic burn-in, and steady state life tests (see 3.6).

b. A copy of each radiograph.

c. The quality conformance inspection data (see 4.4).

d. Parameter distribution data on parameters evaluated during burn-in (see 3.6).

e. Final electrical parmeters data (see 4.2c).

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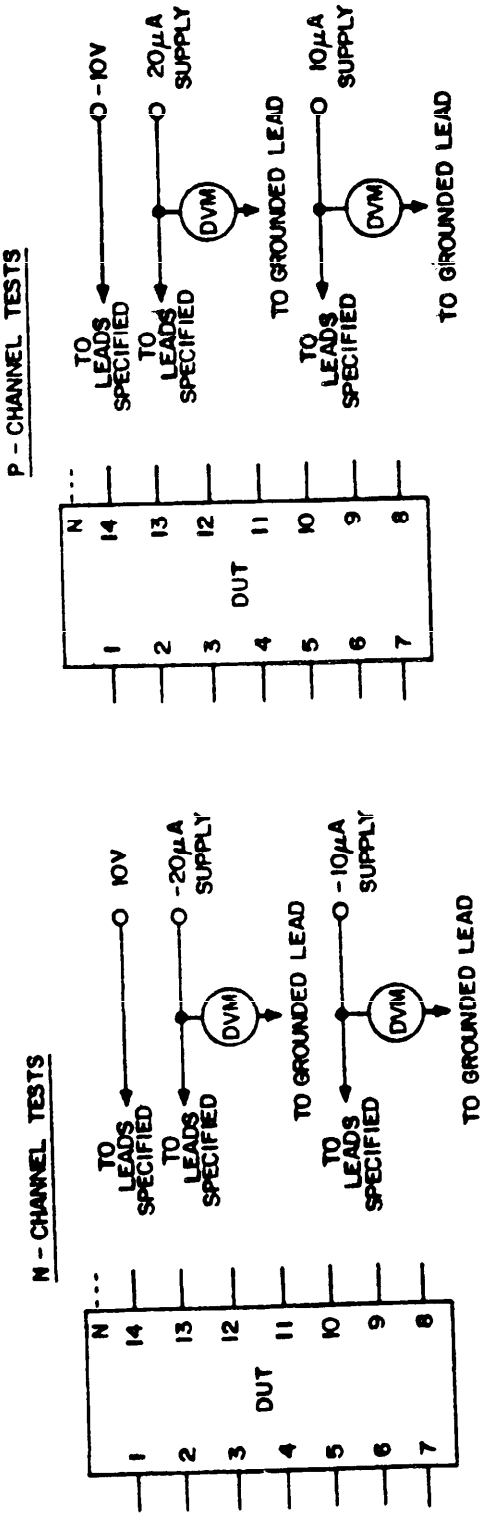


FIGURE 8. Threshold voltage test circuit.

TABLE VIII. Threshold - voltage test circuit conditions.

Device type	GND	10V	V_{TN} measured at		GND	-10V	V_{TP} measured at	
			-20 μA supply	-10 μA supply			20 μA supply	10 μA supply
01	1	14		3,5,7,9,11,13	1	3,5,7,9,11,13		14
02	3	4,10,11,14		7	3	7		4,10,11,14
03	1	3,4,6,10,12, 13,15,16		8	1	8		3,4,6,10,12 13,15,16
04	2	1,3,6,7,9,10, 14,15,16		8	2	8		1,3,6,7,9,10, 14,15,16

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logisitic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for quality assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

i. Requirements for total dose radiation testing (see 3.6.1 and 4.5.5), if applicable.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

V_{ZAP}	Input test voltage.
C_I	Input terminal-to-V_{SS} capacitance.
GND	Ground zero voltage potential.
T_A	Free air temperature.
V_{DD}	Positive supply voltage.
V_{SS}	Negative supply voltage.
I_{SS}	Quiescent supply current.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class S for National Aeronautics and Space Administration or class B for Department of Defense (see 1.2.2), lead finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

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Military device type	Generic-industry type
01	4069UB
02	40107B
03	4502B
04	40109B

6.6 Handling. MOS devices must be handled with certain precautions to avoid damage due to accumulation of static charge. Input protective devices have been designed in the chip to minimize the effect of this static build up. However, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surface.
b. Ground test equipment and tools.
c. Do not handle devices by the leads.
d. Store devices in conductive foam or carriers.
e. Avoid use of plastic, rubber, or silk in MOS areas.
f. Maintain relative humidity above 50 percent, if practical.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - MI
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM, AR
Navy - AS, CG, OC, MC, SH

Agent:
DLA - ES

Preparing activity:
NASA - NA

(Project 5962-0639-5)

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/174A	MICROCIRCUITS, DIGITAL, CMOS, HEX INVERTER, NAND BUFFER DRIVER, STROBED HEX INVERTER?BUFFER, VOLTAGE LEVEL SHIFTER MONOLITHIC, SILICON, POSITIVE LOGIC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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